E-Textiles

Physical Computing(inf175)
Summer Term 2026

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Motivation

Source:Textile Network

Source:Patronus

LED Patches

Exercise Goal:

1. Create a e-textile patch that can be attached to clothing
2. Gain experience with functional materials and soft circuits
3. Learn to deal with unexpected setbacks

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Exercise Task

1. Design an e-textile patch pattern
2. Connect min. 5 LED‘s
3. Use conductive thread
4. Decide on a suitable circuit
5. Sew by hand

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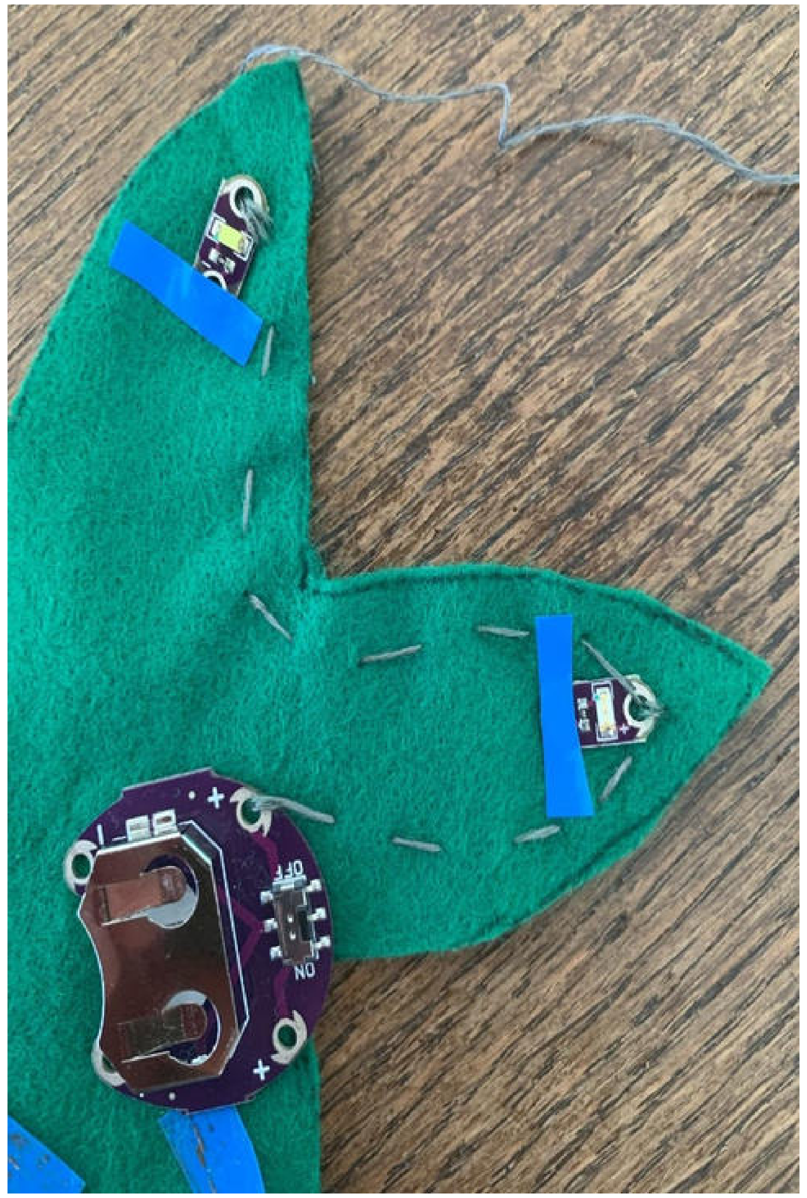
Materials

Source: Brittzay

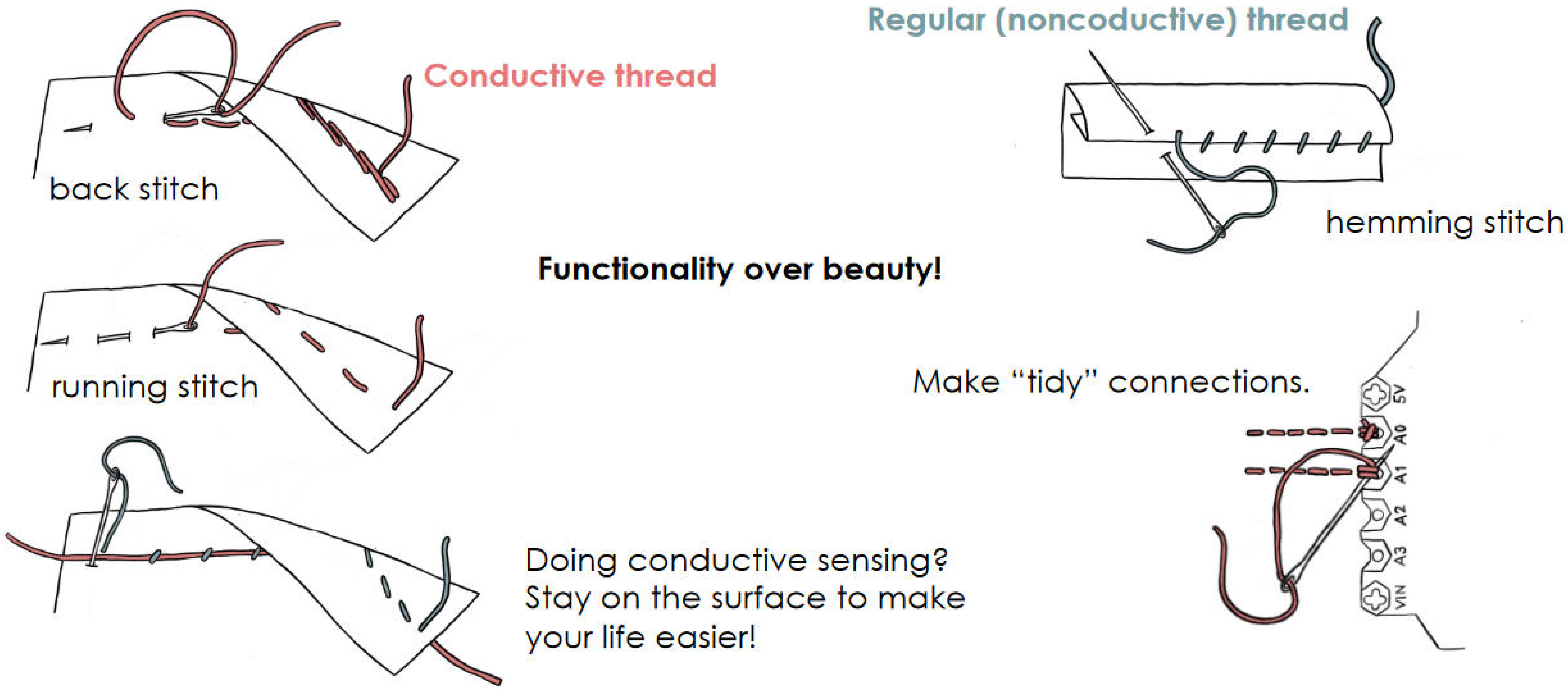
1. Sewable 3V coin battery holder
2. Sewable LED‘s
3. Sewing Needle
4. Conductive Thread
5. Fabric Pen/Chalk
6. Scissors
7. Nail Polish
8. Masking Tape

Getting Started

1. Create your individual pattern
2. Select a base and cover textile
3. Place the battery and LED‘s
4. Connect with conductive thread
5. Attach base to cover piece
6. Attach velcro



Sewing



Picture CC BY Marion Koelle

Circuits

- **Series circuit** has higher resistance
- Yarn also has high resistance
- Not all LEDs will light up
- In a **parallel circuit**, the resistance is divided and the voltage is constant
- All LEDs will light up

Make "proper" connections.

Look out for short circuits!

Mind basic electronic principles., i.e., "the shortest path"

Possible Challenges

Important Notes

E-textile materials (e.g., copper thread) are **very difficult to recycle**. Please use them **economically** and **collect cut-offs and scrap parts** for reuse.

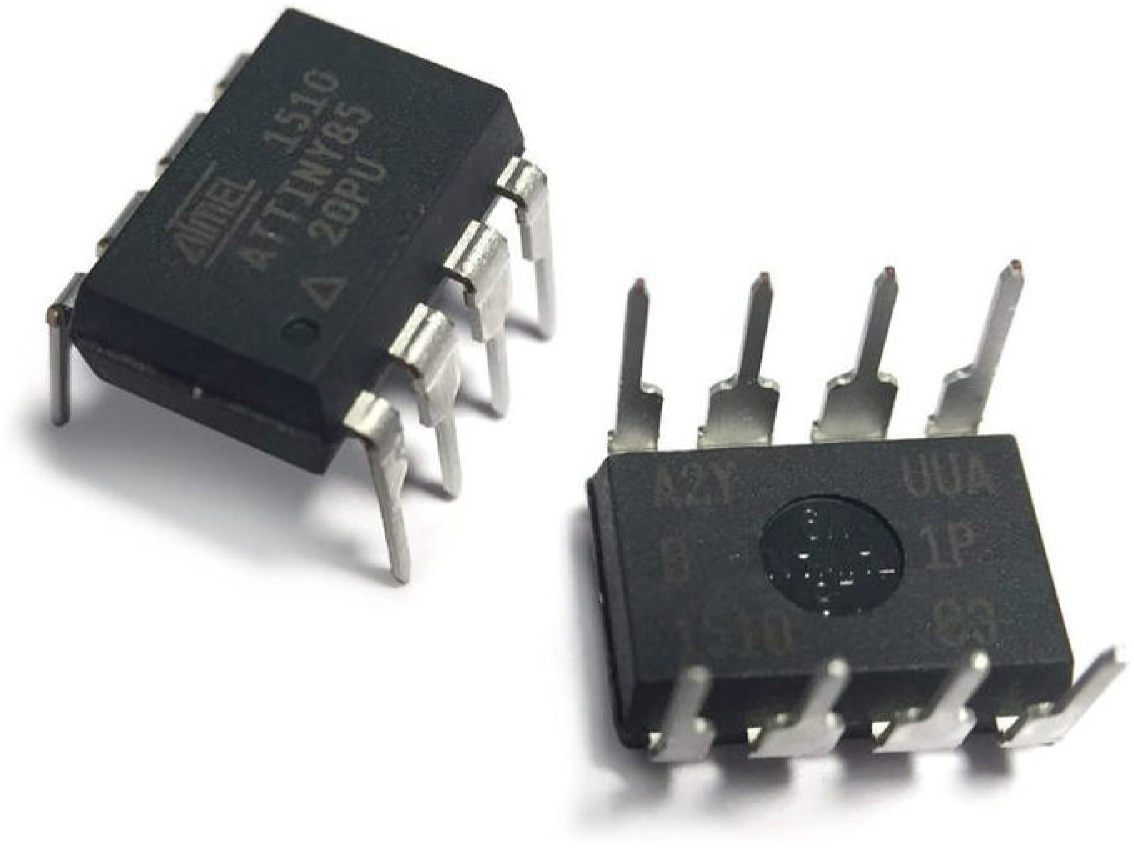
Tips:

- ○ Plan ahead – create a paper template first.
- ○ Measure the exact dimensions and lengths of the materials you need.
- ○ Avoid loosening conductive stitches – test your circuit before sewing.

• **Do not lick conductive thread!** For health and hygiene reasons, always use a needle threader instead.

ATtiny85

- 8-bit Microcontroller
- The smallest Arduino
- Considering the size, it has fewer IO Pins, lower speed, less memory and no USB interface comparing to other Arduino Microcontrollers.
- Attiny 85 has digital PINs, Analog PINs, PWM and SPI PINs and more.
- It's a low power, high-performance, CMOS 8-bit AVR microcontroller chip developed by ATMEL corporation.
- Use case: small and low-power embedded systems.



Attiny85. Source: Electronic Clinic

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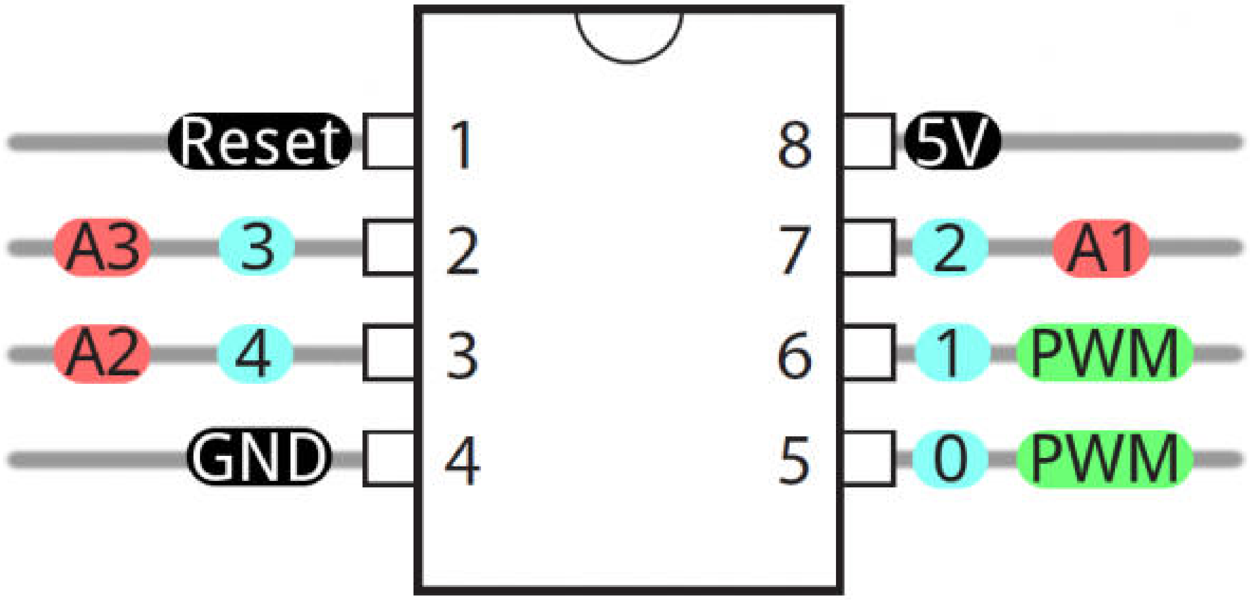
ATtiny85 – technical specifications

- The architecture: AVR(advanced virtual RISC)
- Power supply voltage: from 2.7V to 5.5V.
- CPU speed: up to 20Mhz.
- Flash memory: 8 Kbytes.
- EEPROM is 512 Bytes.
- 6 GPIO and 4 ANALOG pins
- PWM pins are basically located on Pin0 and Pin VM.
- The Interface type is USI(Universal Serial Interfa perform SPI or I2C communications.
- Temperature range is from -40C to +85C



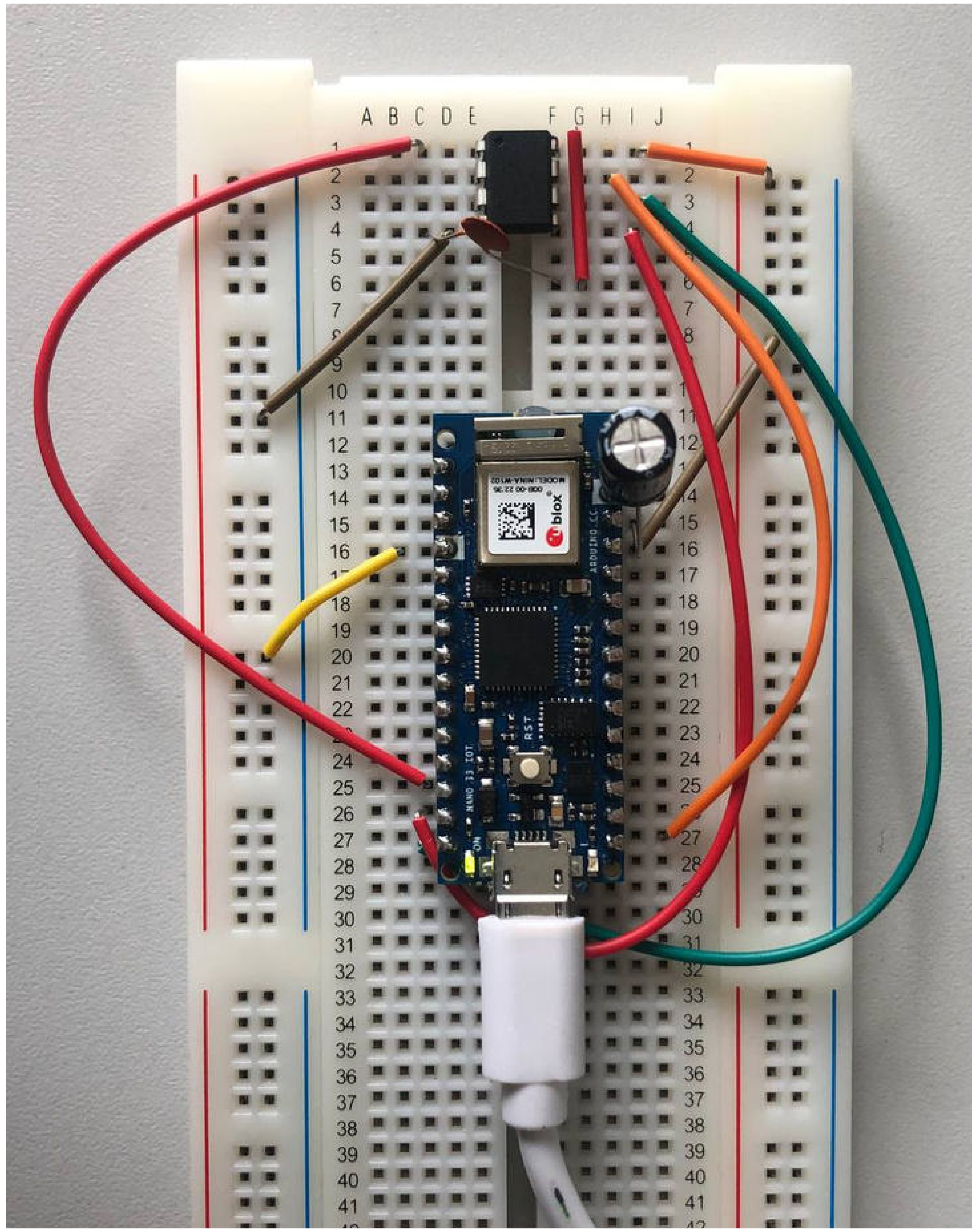
Source: Electronic Clinic

- NO USB interface

Burn the Bootloader

Required components:
- Arduino Nano 33 IoT
- 10k Nano-farad Capacitor
- 104 Capacitor
- ATTiny85
- Jumper Wires

Arduino		ATTINY85
5V		PIN 8
GND		PIN 4
D10		PIN 1
D11		PIN 5
D12		PIN 6
D13		PIN 7



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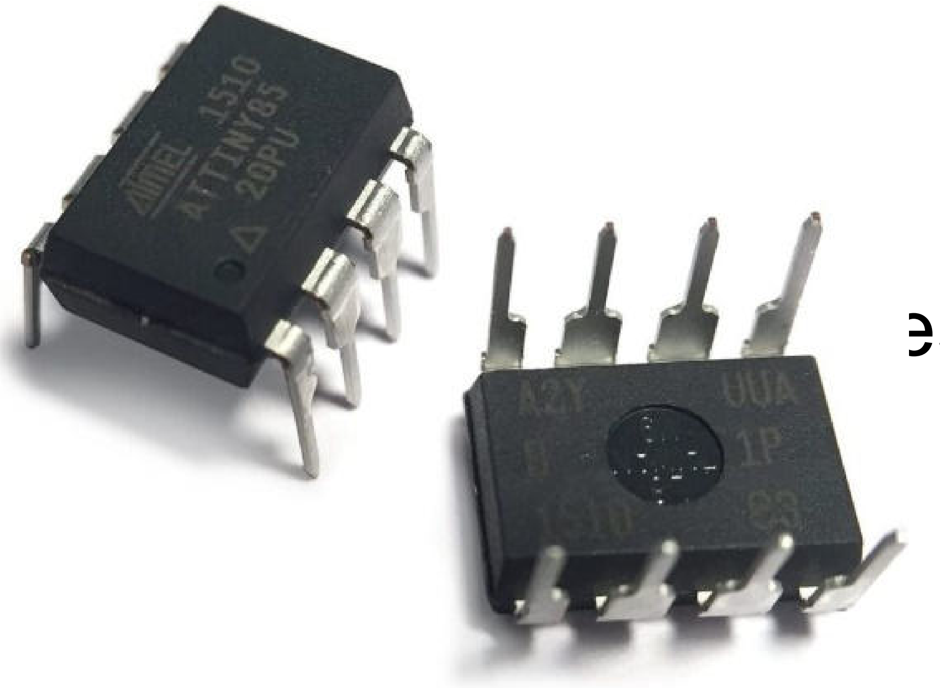
Burn the Bootloader

1. Connect the PIN1 of ATtiny85 which is also the reset pin, to pin10 of the Arduino which is the SS PIN (Slave Select).
2. Connect ground pins (PIN 4 of Attiny to GND of Arduino)
3. Connect pin 5 (MOSI PIN) to pin 11 of the Arduino (Master out, slave in)
4. Connect pin 6 (MISO) to pin 12 of Arduino.
5. Connect pin 7 (SCK) to pin 13 of the Arduino, which is the SCK (Serial Clock)
6. Connect pin 8 which is the VCC pin to the Arduino 5V pin.
7. Connect a 10uF capacitor between the Reset and GND pins of Arduino (positive leg of the capacitor to reset pin and negative leg to the Ground.

Arduino		ATTINY85
5V		PIN 8
GND		PIN 4
D10		PIN 1
D11		PIN 5
D12		PIN 6
D13		PIN 7

Installing the ATTiny Core



Since the ATtiny85 packages are not preinstallec essary to manually install it:

- Open Arduino IDE
- Go to files-> preferences
- Enter this link in the preferences menu: ATtiny85 packages
- Then go to tools-> Board and click “Board
- Search ATTiny and install “ATTiny” [Versi
- Click Tools-> Boards-> Attiny and select tl

Arduino	ATTINY85
5V	PIN 8
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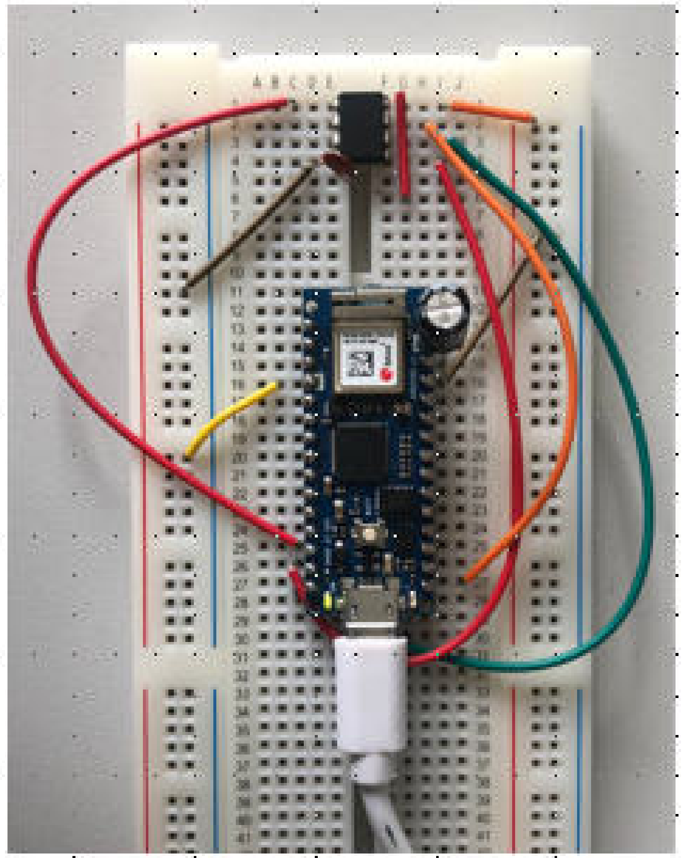
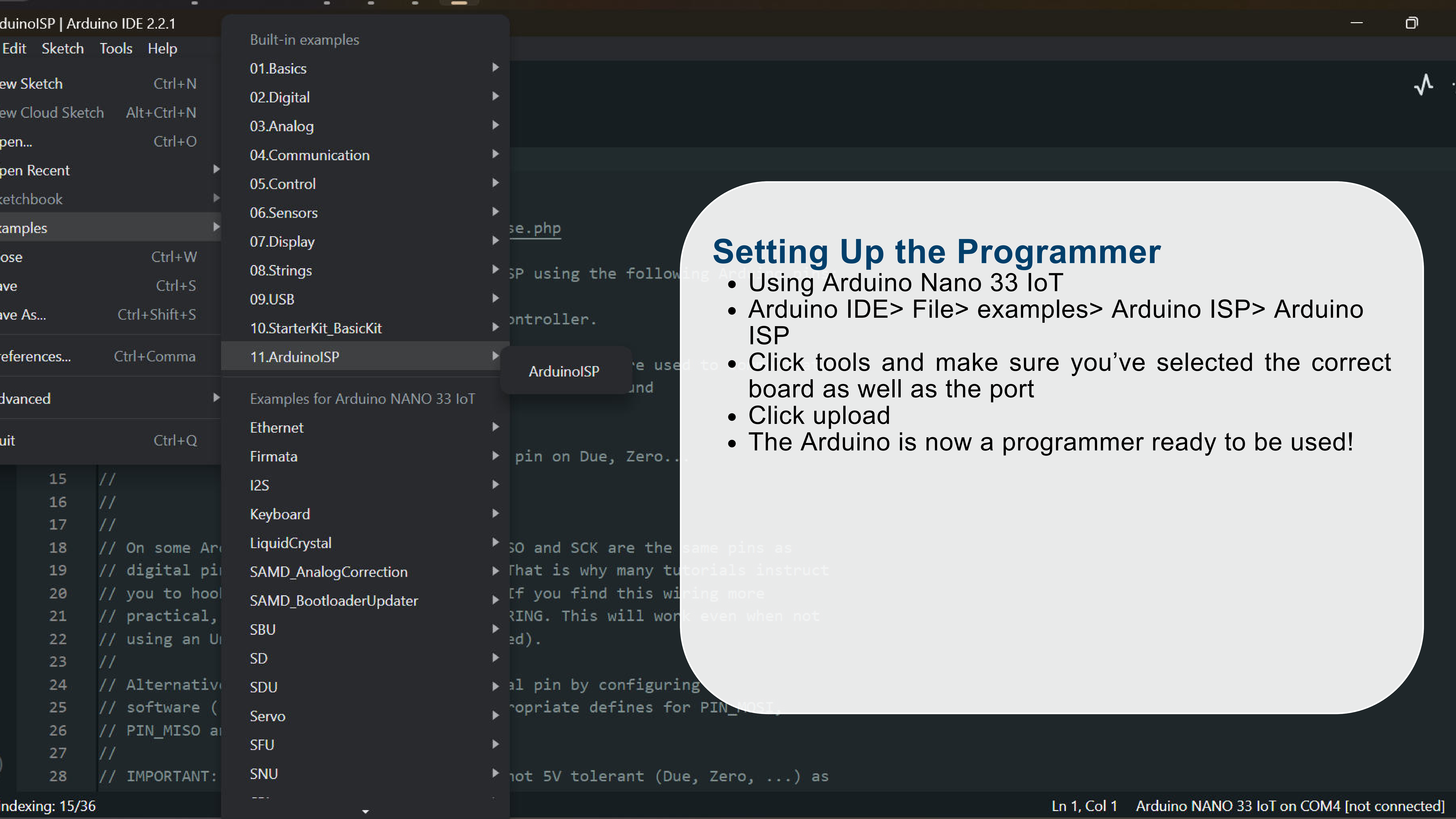


Complete the Bootloader burning process

- Next step would be to select tools> programmer, then set the programmer to “Arduino as ISP”
- Finally select Run Bootloader from Tools menu.
- After the process is completed, there is a “Done burning bootloader” message in the IDE.

Setting Up the Programmer

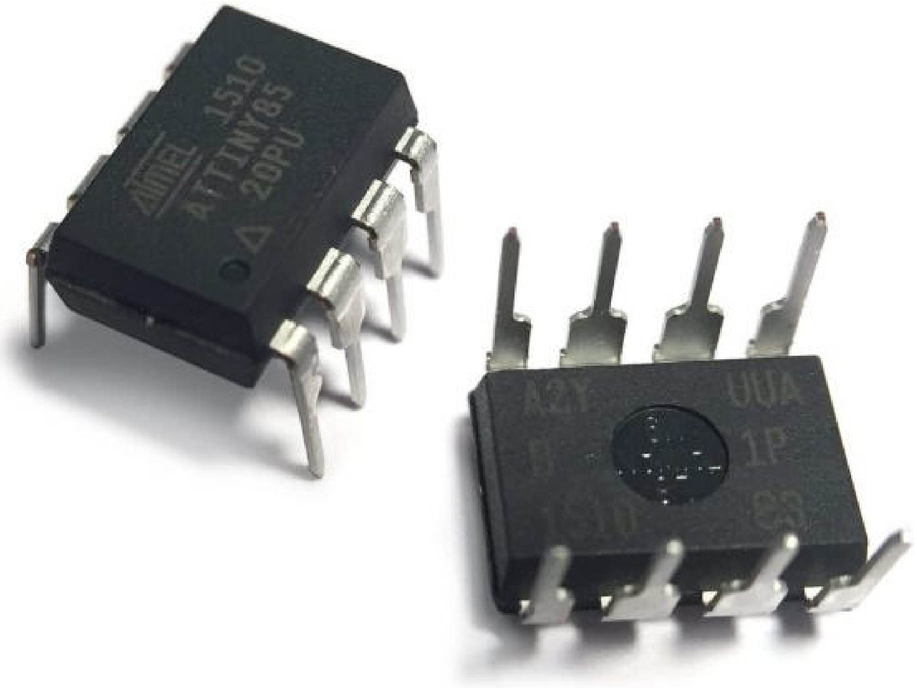
- Using Arduino Nano 33 IoT
- Arduino IDE> File> examples> Arduino ISP> Arduino ISP
- Click tools and make sure you've selected the correct board as well as the port
- Click upload
- The Arduino is now a programmer ready to be used!



Program ATtiny85 using the Arduino IDE

- Set up the circuit, in this example there is an LED to be controlled.
- Connect the power source and ground wires with the VCC and GND pins of ATtiny85 microcontroller.
- Connect the Anode link of the LED to pin 5 which is the digital pin 0, so in programming we will refer to the physical pin 5 is pin 0.
- Connect the other leg of the LED to the ground and a 330Ohms Resistor in between.
- Done!

Extra Task (for bonus points)



1. Use an ATtiny (8-bit Microcontroller)
2. Burn the bootloader
3. Install ATtiny85 package in Arduino IDE
4. Program ATtiny85 that controls the LED‘s
5. Connect ATtiny to your e-textile patch

Arduino	ATTINY85
5V	PIN 8
GND	PIN 4
D10	PIN 1
D11	PIN 5
D12	PIN 6
D13	PIN 7